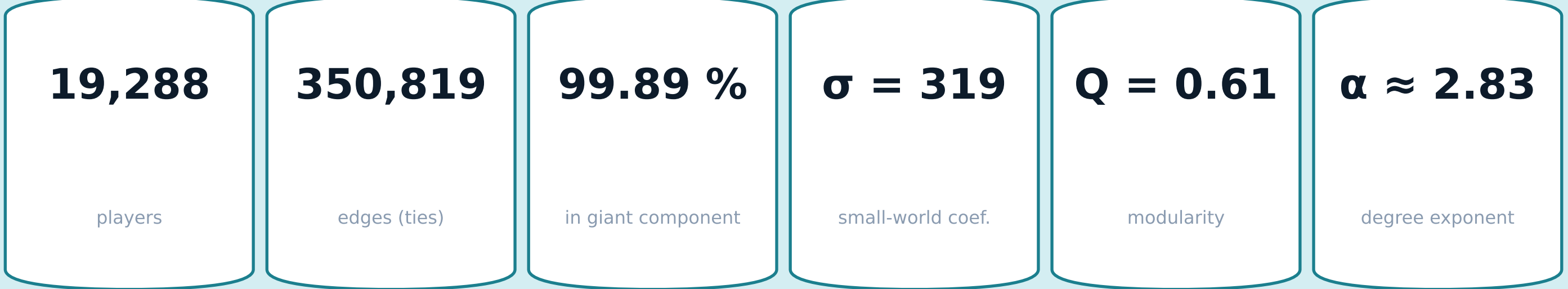


HEADLINE FINDING

The transfer network is a textbook small-world.

370× more clustered than random — yet any two players are ≤ 7 hops apart.



01 · INTRODUCTION

Football's transfer market is one of the few global labour markets where every job change is publicly recorded for 30+ years. We model it as a complex network: each player is a node, and an edge connects two players who shared the same club in the same season — a co-tenancy tie. Edge weights count shared seasons.

This co-membership graph lets us ask: how clustered are squads, how short are career paths, did the 1995 Bosman ruling leave a topological fingerprint, and can structural features predict the next career overlap?

02 · DATA

Source: ewenme/transfers (public GitHub CSV, MIT).
Leagues: Premier League · La Liga · Bundesliga · Serie A · Ligue 1.
Seasons: 1992 → 2024 (33 seasons).
Records: 46,674 incoming transfers after de-duplication.

Per row we keep: player · club · season · age · position · fee · transfer movement · league · country. Out-rows are dropped because every transfer is recorded twice — once by the receiver and once by the seller.

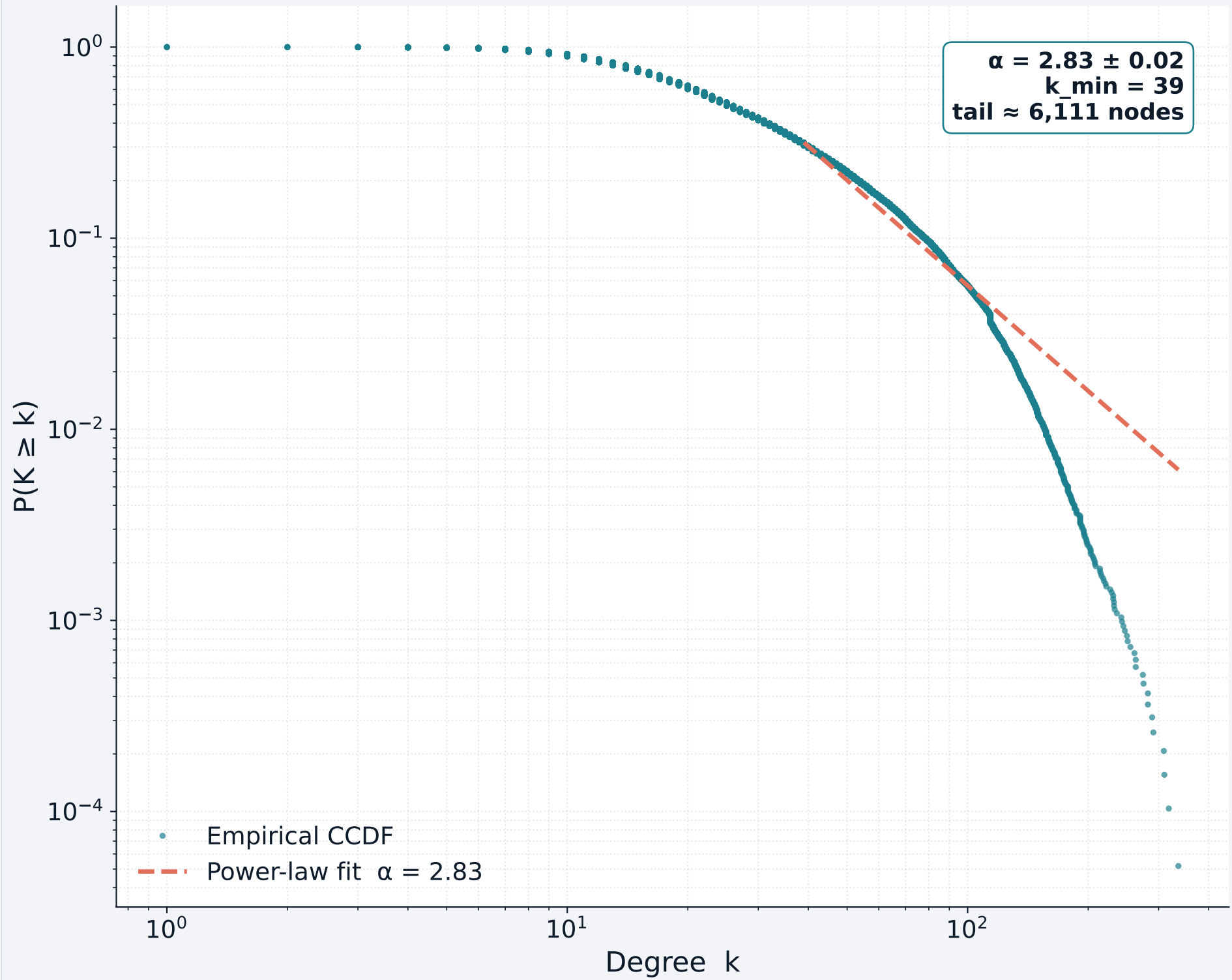
03 · METHOD

Bipartite construction. Build a player ↔ (club, season) bipartite graph, then project onto the player axis. Two players link if they shared at least one (club, season); edge weight counts shared cells.

Analyses. Power-law fit (Clauset-Shalizi-Newman), five centralities, Louvain communities, Watts–Strogatz small-world coefficients σ and ω , k-core, assortativity, temporal snapshots, and a logistic-regression link-prediction ensemble.

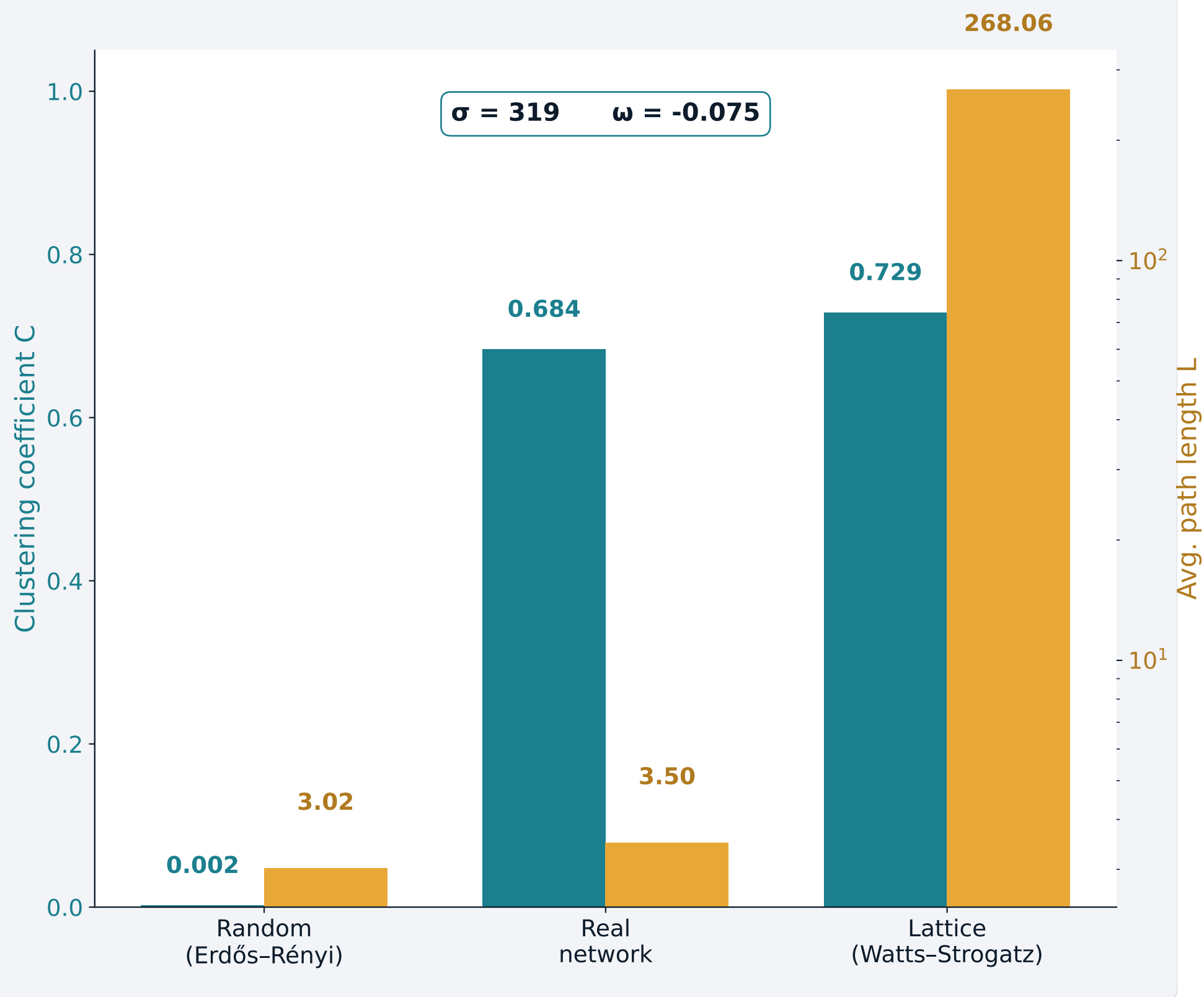
04 · DEGREE DISTRIBUTION

A few super-connected journeymen — most players have only a handful of co-tenants.



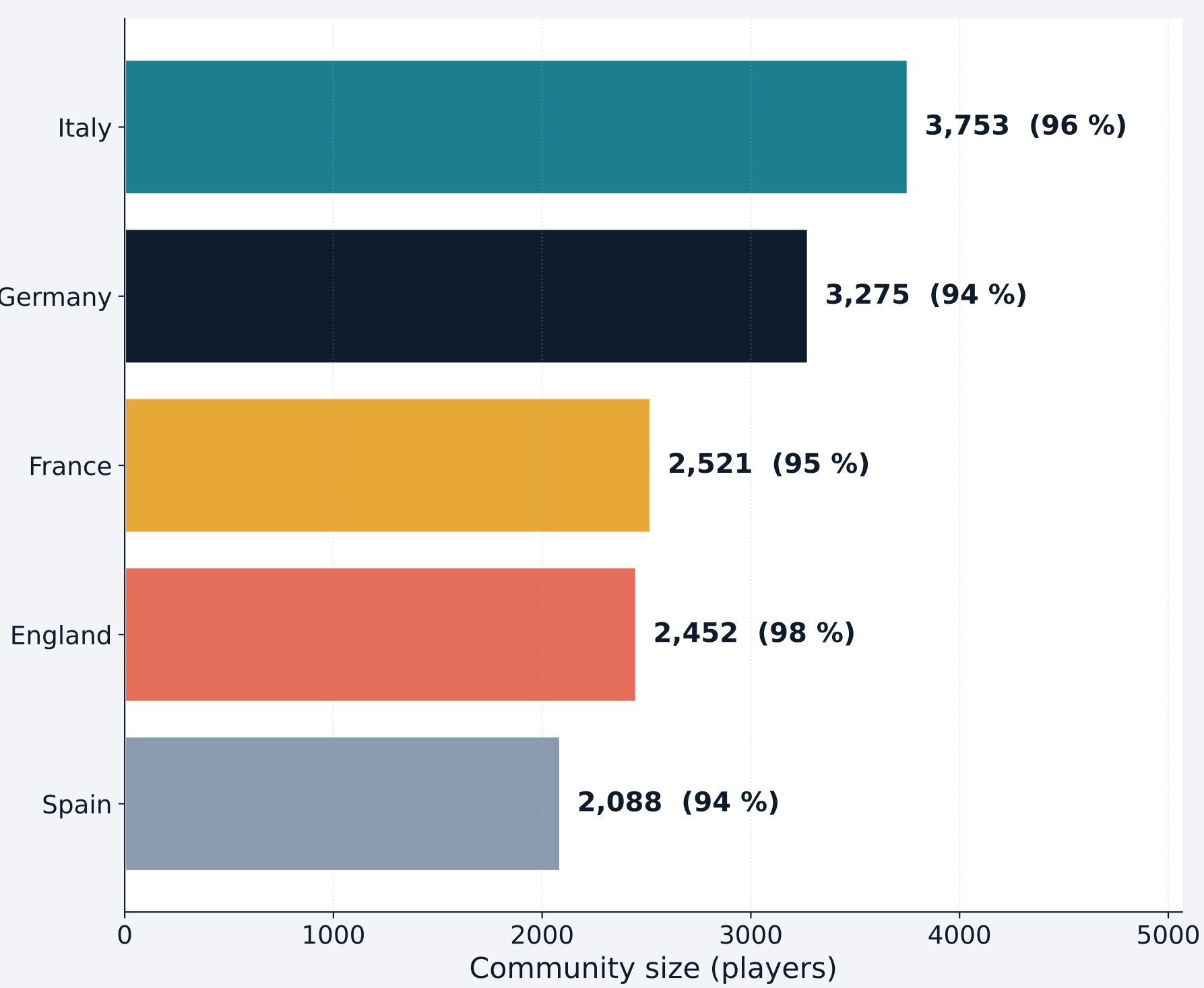
05 · SMALL-WORLDNESS

Lattice-like clustering + random-graph-like path length = small world.



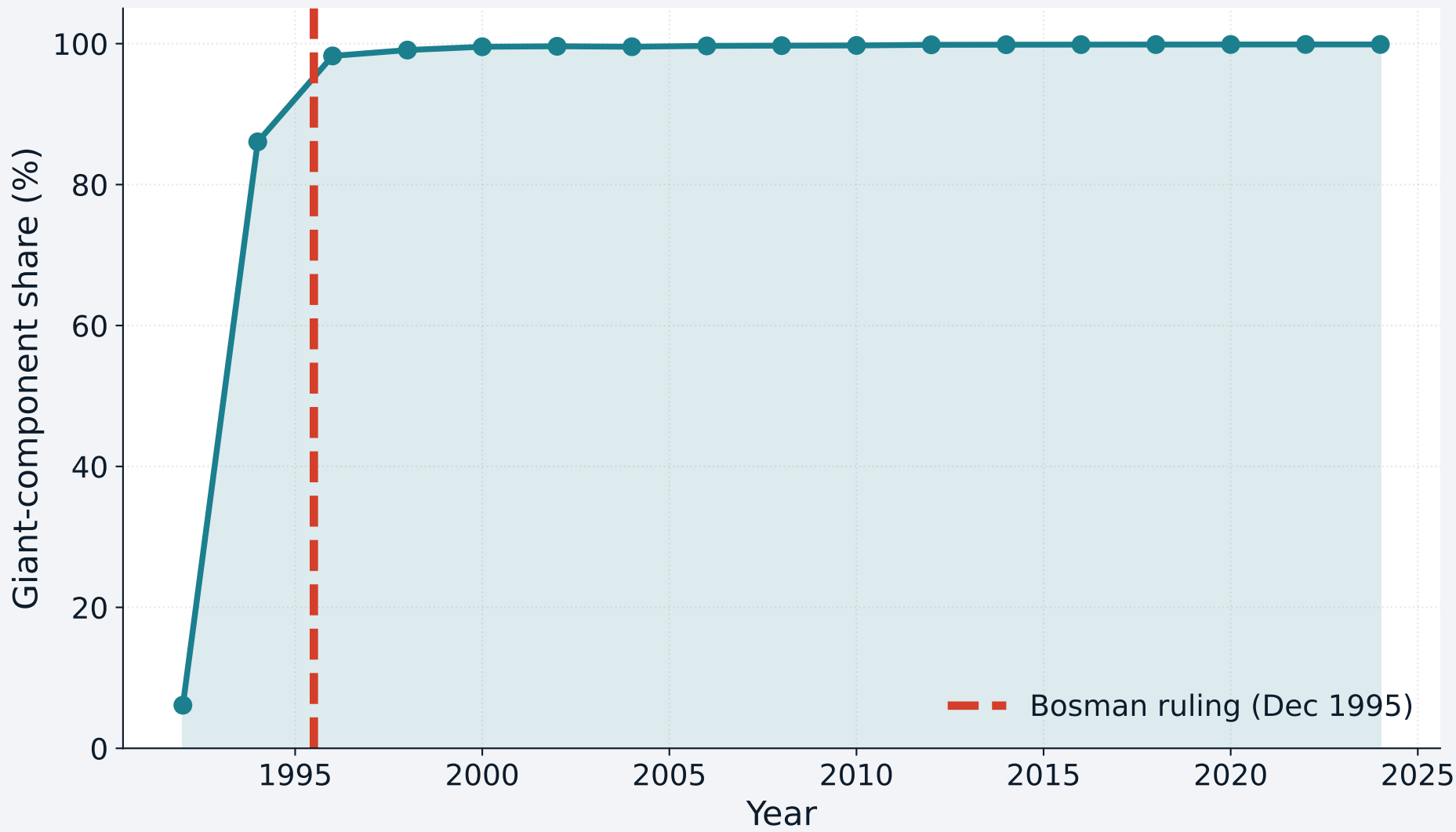
06 · COMMUNITIES

21 communities, $Q = 0.61$. The five biggest are each ≥ 94 % single-country.

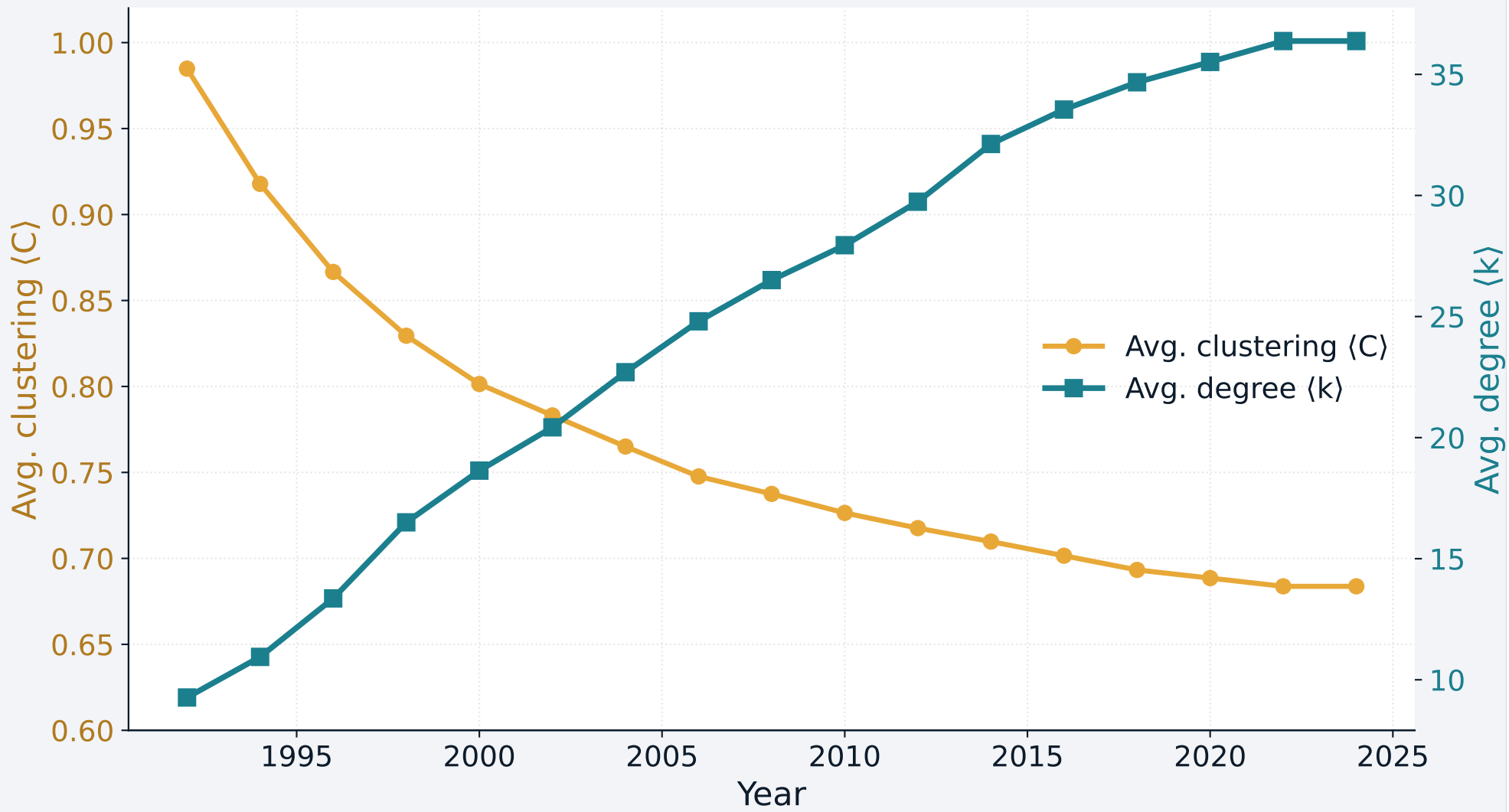


07 · TEMPORAL DYNAMICS — THE BOSMAN PHASE TRANSITION

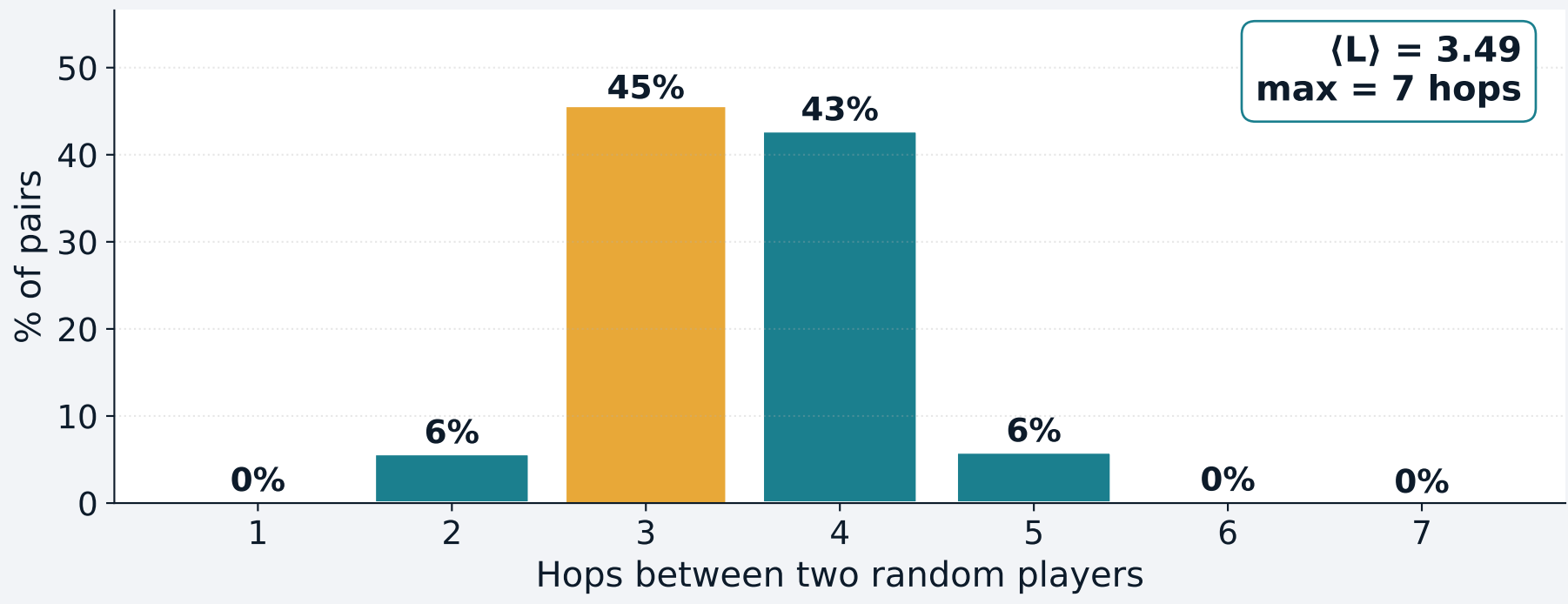
Belgian midfielder Jean-Marc Bosman sued his old club for blocking his free move after his contract expired — in 1995 the EU ruled in his favour: out-of-contract EU players can now switch clubs as free agents, no transfer fee. Our giant component (largest reachable cluster) explodes from 6 % (1992) to 98 % (1996), straddling that ruling exactly.



Clustering decays as the network grows (early rosters are forced cliques); avg degree quadruples ($9 \rightarrow 36$) and density drops $6\times$.



08 · SIX DEGREES



Famous journeys

- CR7 ≠ Pirlo: 2 hops (via M. Cáceres)
- Mbappé ≠ farthest: 7 hops
- Pairs reachable ≤ 6 hops: 99.998 %

Everyone in elite European football is a handful of co-tenancy hops from any other player.

09 · WHO'S CENTRAL?

Journeymen, not superstars, top every centrality — the players who hop between clubs are the structural backbone.

DEGREE

1. Felipe 2. Antonio Cassano 3. Gaël Kakuta

BETWEENNESS

1. Fernando 2. Júlio César 3. Gaël Kakuta

PAGERANK

1. Fernando 2. Gaël Kakuta 3. Marco Borriello

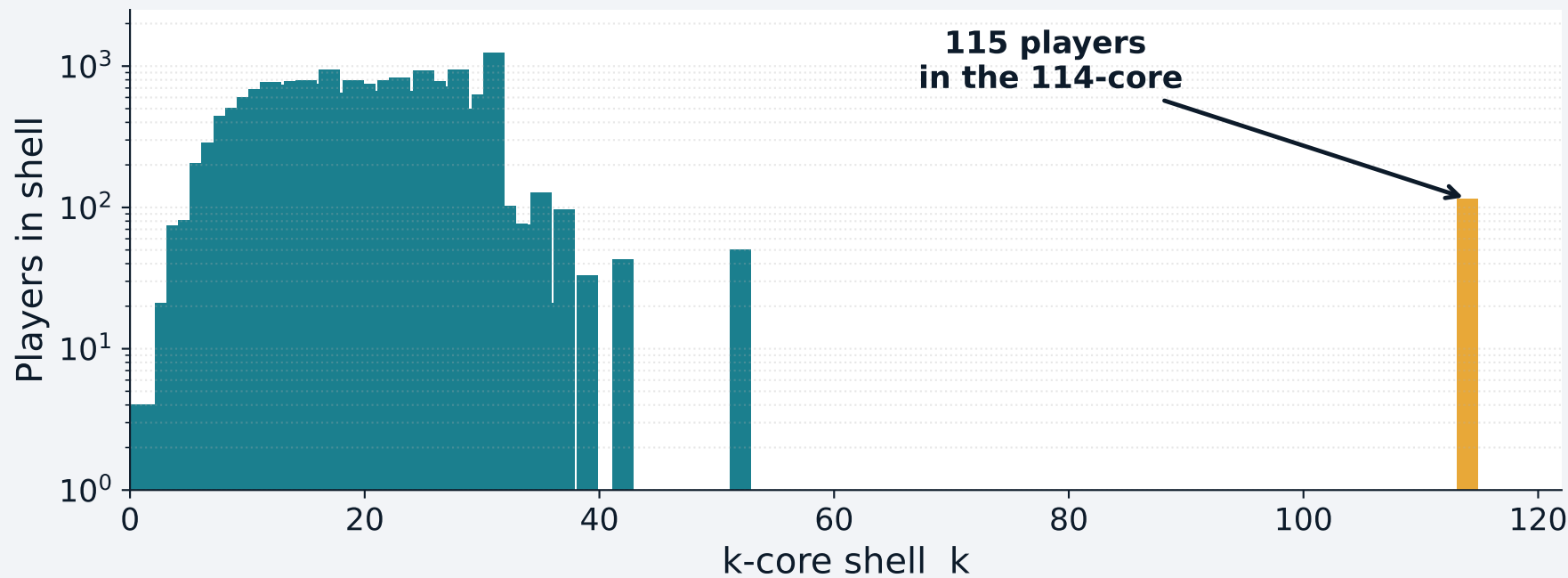
EIGENVECTOR

1. Antonio Cassano 2. Mattia Cassani 3. Ezequiel Schelotto

Lusophone single-names (Felipe, Fernando) may merge in the source — see limits.

10 · THE 114-CORE: INNER 115

115 players form a tight inner ring — every pair shares at least 114 mutual co-tenants. The journeyman elite of European football.



Members — classic journeymen:

Cassano · Kakuta · Borriello · Eto'o · Boateng · ...

A 114-core is the densest k-core — it quantifies the "inner ring" of the network.

11 · TOPOLOGY SNAPSHOT

Density	1.89e-03
Triangles	2,429,012
Transitivity (global C)	0.324
Avg. local clustering	0.684
C / C_random	370 ×
Diameter (BFS, k=500)	≥ 7
Radius	≤ 5
Degree assortativity r	+0.191
Max k-core	$k = 114$

Positive $r \Rightarrow$ hubs co-locate with hubs — elite-roster effect.

12 · TAKEAWAYS

► One ecosystem.

99.89 % of all 19,288 elite-five-leagues players sit in a single connected component — one labour-market graph.

► Textbook small-world.

$C = 0.68$ ($370\times$ random) and $L = 3.5$ ($1.16\times$ random) give $\sigma \approx 319$ and $\omega \approx 0$ — clustering of a lattice, reach of a random graph.

► Bosman is visible.

GCC jumps 6 % → 98 % between 1992 and 1996, straddling the 1995 ruling — regulation reshapes topology.

► Six degrees of football.

Mean path 3.49, max 7. The classic six-degrees phenomenon holds at the scale of European football.

► Communities trace borders.

21 Louvain communities ($Q = 0.61$); the five largest are each ≥ 94 % single-country — structurally national market.

► A 115-player inner ring.

The densest k-core ($k = 114$): every pair shares ≥ 114 mutual co-tenants — football's structural elite.

13 · INTERESTING FINDINGS

► Six degrees, for real.

99.998 % of all —186 million player pairs are reachable in 6 hops or fewer.

► Squads travel together.

2.4 million triangles — co-tenants of co-tenants are usually co-tenants too.

► Fernando is the backbone.

#1 in both betweenness AND PageRank — the single most critical bridge.

► Star paths are short.

Cristiano Ronaldo to Andrea Pirlo: 2 hops via Martín Cáceres.

► Country mixing is intense.

Assortativity $r = +0.74$ — far above social-network norms (~ -0.2).

► 2010s, football's busiest decade.

Over 150 k new edges added — more than any prior decade.